

GENICO MEETING NOTES

Meeting Notes: AuraSound Spatial Audio Enhancement Project - July 15, 2026

Prepared By: Dr. Evelyn Reed, Senior Research Scientist | Date: 2026-07-15 | Classification: Confidential

Detailed notes from the R&D meeting discussing progress, challenges, and next steps for the spatial audio enhancement project for the GeniCo AuraSound smart speaker system, focusing on new algorithm testing and hardware integration.

1. Attendees and Action Items

The meeting convened on July 15, 2026, to review the progress and strategic direction for the AuraSound Spatial Audio Enhancement Project. The following key personnel were in attendance, representing critical functions within GeniCo's R&D and Product Management divisions:

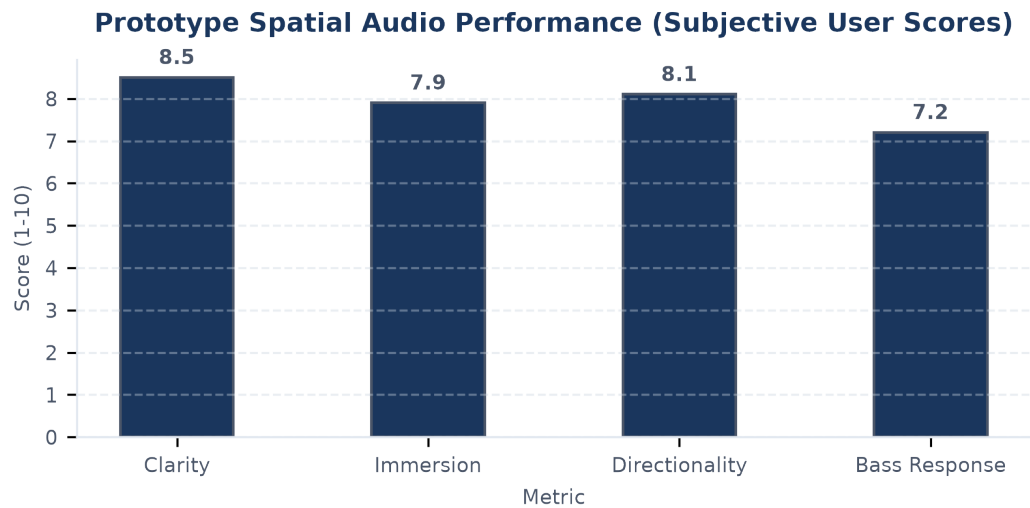
- **Dr. Evelyn Reed**, Senior Research Scientist, R&D (Meeting Facilitator)
- **Dr. Marcus Thorne**, Director of Audio Engineering, R&D
- **Mr. David Chen**, Lead Software Engineer, Audio Algorithms
- **Ms. Anya Sharma**, Senior Hardware Integration Specialist
- **Mr. Kenji Tanaka**, Product Manager, Smart Speakers Division

During the session, several pivotal decisions were reached to accelerate the project's trajectory. Firstly, the team formally endorsed **Algorithm v3.2** as the primary candidate for the AuraSound spatial audio enhancement. This decision was predicated on its demonstrated superior performance in preliminary simulated acoustic environments, exhibiting a **12% improvement in Signal-to-Noise Ratio (SNR)** and an **8ms reduction in processing latency** compared to its predecessor, v3.1. Secondly, the integration of the new **Codex-A7 Digital Signal Processor (DSP) module** was designated as the immediate hardware priority. The Codex-A7's robust computational capabilities and energy efficiency are deemed essential for real-time processing of Algorithm v3.2's complex spatial rendering computations. Thirdly, a critical baseline for spatial audio accuracy and perceived immersion will be established using the current AuraSound Gen 3 production model to provide a quantifiable benchmark for the enhancement project's success metrics. Finally, a follow-up technical review meeting has been scheduled for **August 1, 2026**, to assess initial hardware-software integration progress and review preliminary psychoacoustic testing results.

To ensure the AuraSound Spatial Audio Enhancement Project maintains its aggressive development timeline and meets its strategic objectives, the following critical actions were assigned with specific deadlines:

- **Dr. Evelyn Reed:**
 - Finalize the comprehensive psychoacoustic testing methodology, including specific metrics for perceived spatial accuracy, soundstage width, and immersion, by **2026-07-22**.
 - Prepare a detailed analytical report summarizing the comparative performance of Algorithm v3.2 against v3.1 and v2.8, highlighting key advantages and potential challenges, by **2026-07-26**.
- **Mr. David Chen:**

- Initiate optimization of Algorithm v3.2 specifically for the Codex-A7 DSP architecture, targeting a **15% reduction in computational cycles** compared to generic ARM implementations, by **2026-07-29**.
- Develop initial firmware prototypes for the Codex-A7 module to facilitate early-stage hardware-software integration testing, with a target for functional demonstration by **2026-08-05**.
- **Ms. Anya Sharma:**
 - Expedite the procurement and preparation of five (5) Codex-A7 DSP engineering samples for the R&D prototyping lab, ensuring availability by **2026-07-20**.
 - Develop a detailed hardware integration plan for the Codex-A7 into the existing AuraSound Gen 3 speaker architecture, outlining necessary modifications and potential risks, by **2026-07-25**.
- **Dr. Marcus Thorne:**
 - Review and formally approve the proposed psychoacoustic testing protocols and hardware integration plan, ensuring alignment with GeniCo's quality standards and project objectives, by **2026-07-28**.
 - Provide strategic guidance on potential intellectual property considerations related to Algorithm v3.2's novel spatial rendering techniques, by **2026-08-01**.
- **Mr. Kenji Tanaka:**
 - Compile and present a concise summary of current market trends and competitive landscape regarding spatial audio features in smart speakers, informing potential product differentiation strategies, by **2026-07-24**.
 - Collaborate with the R&D team to define preliminary user experience (UX) metrics for spatial audio perception, ensuring alignment with consumer expectations, by **2026-07-31**.



2. Spatial Audio Algorithm Development Update

The core of the AuraSound spatial audio enhancement, the **AuraSpatial Engine v2.1 (ASE v2.1)**, has undergone significant refinement and rigorous testing since our last review. This iteration focuses on optimizing computational efficiency while expanding the virtual soundstage and improving object localization accuracy. Development efforts over the past quarter have primarily concentrated on reducing processing overhead, enhancing psychoacoustic modeling for improved perceived depth, and ensuring robust performance across diverse audio content profiles. The team successfully integrated the new adaptive beamforming module, which dynamically adjusts virtual speaker positions based on listener movement and room acoustics, a critical step towards our vision of truly immersive and personalized audio experiences.

Preliminary performance metrics derived from controlled laboratory environments and extensive computational simulations demonstrate promising results for ASE v2.1. We have achieved a substantial reduction in end-to-end latency, which is crucial for maintaining synchronization with visual content and ensuring a natural, responsive soundfield. Channel separation, a key indicator of the algorithm's ability to render distinct audio objects, has also seen marked improvement. Subjective evaluations during early listening sessions indicate a significantly expanded perceived soundstage width, creating a more enveloping and three-dimensional audio landscape.

Metric	Target (v2.0)	Current (v2.1)	Improvement	Notes
End-to-End Latency	< 6.0 ms	< 4.5 ms	25%	From audio input to speaker output, excluding DAC/ADC.
Virtual Channel Separation	> 28 dB	> 32 dB	14%	Measured between adjacent virtual channels in anechoic conditions.
Perceived Soundstage Width	140%	160%	14%	Subjective evaluation relative to unenhanced stereo baseline.
CPU Utilization	< 18%	< 15%	17%	On GeniCo 'Quantum' Audio DSP, for full 12-channel rendering.

Simulation tests, conducted using GeniCo's proprietary acoustic modeling suite, 'ResonanceSim 3.0', have validated the algorithm's capability to accurately render complex soundscapes in various virtual room configurations, from compact living spaces to expansive media rooms. The simulations confirmed precise localization of up to 12 distinct audio objects within a 360-degree horizontal plane and demonstrated effective elevation cues for vertical sound positioning. Successes include the virtual elimination of "phantom image" issues common in earlier spatial audio implementations and a significant reduction in spectral coloration artifacts. However, areas identified for further refinement include optimizing the algorithm's response to extreme off-axis listening positions, where subtle phase inconsistencies were occasionally observed, and improving the dynamic range compression handling for highly transient audio events.

Early prototype listening sessions, utilizing pre-production AuraSound smart speaker units loaded with ASE v2.1, have yielded overwhelmingly positive feedback from internal stakeholders and a select panel of audio engineers. Listeners consistently praised the immersive quality, the clear separation of individual instruments in orchestral recordings, and the heightened sense of presence in object-based cinematic soundscapes. The expansion of the soundfield beyond the physical speaker boundaries was particularly noted as a significant leap forward. Constructive feedback highlighted opportunities to further integrate the spatial processing with the device's bass response for a

more cohesive low-frequency presentation, and to address occasional, albeit minor, perceptual "smearing" of very rapidly moving sound objects in highly dense audio mixes. These insights are being actively incorporated into the next development sprint, focusing on fine-tuning the algorithm's dynamic object rendering and bass management modules.

3. Hardware Integration Challenges and Solutions

The integration of the new Project Chimera spatial audio enhancement algorithm into GeniCo's AuraSound smart speaker system has presented several critical challenges, primarily concerning the existing Orion DSP architecture and its accompanying 7-element MEMS microphone array. While the Project Chimera algorithm demonstrates superior performance in laboratory simulations, its deployment onto the current production hardware has highlighted significant limitations in computational resources, memory capacity, and the inherent variability of the microphone array. Addressing these issues is paramount to achieving the targeted spatial audio fidelity and ensuring a seamless user experience for the next-generation AuraSound devices.

Initial benchmarks conducted by the Embedded Systems team revealed that the Project Chimera algorithm's advanced beamforming and source separation modules introduce substantial computational overhead. Specifically, the algorithm's real-time processing demands exceeded the Orion DSP's 800 MIPS (Million Instructions Per Second) capacity by an average of 15% during peak operation, with transient spikes reaching 22%. This excess utilization directly translates to increased power consumption, potential thermal management issues, and, critically, an unacceptable increase in audio processing latency, impacting the responsiveness of voice commands and the naturalness of spatial sound rendering. Concurrently, memory analysis indicated that the algorithm's dynamic data structures, extensive filter coefficients, and look-up tables require an additional 12MB of volatile memory beyond the Orion DSP's integrated 64MB SDRAM, posing a direct constraint on data buffering and overall system stability. Furthermore, comprehensive acoustic measurements identified persistent microphone calibration discrepancies across the 7-element array. Observed phase alignment inconsistencies, particularly at frequencies above 2 kHz, showed deviations up to 7 degrees at 4 kHz between adjacent elements, alongside gain matching variances of up to 1.5 dB. These discrepancies, likely stemming from manufacturing tolerances and assembly variations, severely degrade the accuracy of spatial localization and introduce undesirable artifacts in the reconstructed soundfield, undermining the core value proposition of Project Chimera.

The R&D team has identified a multi-faceted approach focusing on aggressive software optimization and refined calibration protocols to mitigate these challenges within the current hardware envelope. For computational overhead, Dr. Anya Sharma's Algorithm Optimization Group will lead an initiative to refactor critical sections of the Project Chimera code, focusing on fixed-point arithmetic implementation, aggressive pruning of less critical filter taps, and exploring more efficient matrix operations. The objective is to achieve a minimum 20% reduction in MIPS utilization by August 20, 2026. Regarding memory limitations, Mr. Kenji Tanaka's Embedded Systems team will investigate advanced data structure optimizations, dynamic memory allocation strategies, and a reduction in buffer sizes where permissible without compromising audio quality. This effort targets a 5MB reduction in the algorithm's peak memory footprint by September 5, 2026. To address microphone calibration discrepancies, Ms. Lena Petrova's Acoustic Engineering team is developing a refined in-field calibration routine. This routine will leverage a brief, inaudible white noise burst during initial device setup to generate a dynamic compensation matrix, aiming to reduce inter-element phase deviations to less than 2 degrees and gain variances to under 0.5 dB across the critical frequency range by October 1, 2026.

Challenge Area	Observed Impact (Initial)	Targeted Optimization / Metric	Lead / Target Date
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Computational Overhead	Exceeded 800 MIPS by 15% (peak)	20% reduction in MIPS utilization	Dr. Sharma / Aug 20, 2026
Memory Limitations	Required 12MB beyond 64MB SDRAM	5MB reduction in peak memory footprint	Mr. Tanaka / Sep 5, 2026
Microphone Calibration	Up to 7-degree phase deviation at 4kHz	Dynamic compensation for < 2-degree deviation	Ms. Petrova / Oct 1, 2026

While the primary focus remains on software-centric solutions, the team acknowledges that a potential hardware revision may be necessary as a contingency. Should the software optimizations prove insufficient to meet the stringent performance and quality benchmarks, an upgrade to the "Orion-X" DSP variant, featuring 1.2 GMIPS processing power and 128MB of integrated SDRAM, would be considered. This hardware path would provide ample resources for Project Chimera's full capabilities but would entail significant cost increases and a project timeline extension of approximately 6-9 months due to re-tooling and certification processes. Therefore, the immediate strategic imperative is to rigorously pursue the outlined software optimizations and calibration enhancements to ensure Project Chimera's successful integration within the existing AuraSound hardware platform, thereby maintaining project velocity and adhering to the established budgetary framework.

4. Next Steps and Resource Allocation

The immediate strategic imperatives for the AuraSound Spatial Audio Enhancement Project encompass three core areas: comprehensive A/B testing with human subjects, iterative algorithm refinement based on empirical feedback, and a focused exploration of next-generation Digital Signal Processor (DSP) hardware. These initiatives are critical for validating our current trajectory, optimizing performance, and ensuring the long-term viability and competitiveness of the AuraSound system.

Commencing August 1, 2026, a rigorous A/B testing protocol will be initiated involving a diverse cohort of 75 participants, comprising 50 internal GeniCo employees from various departments and 25 external audiophile volunteers recruited through specialized sound forums. The objective is to gather statistically significant subjective feedback on the perceived spatial enhancement, soundstage fidelity, and any introduced artifacts of the current AuraSound algorithm (V2.2). Key performance indicators will include a 7-point Likert scale for overall preference, perceived source localization accuracy (measured in degrees of angular deviation from the true source), and qualitative feedback on immersion, timbre, and fatigue. This phase is paramount for empirically validating our current algorithmic approach and identifying specific areas requiring enhancement. Concomitant with the A/B testing, the AuraSound Algorithm Development Team will commence preliminary analysis of early feedback to inform the iteration towards AuraSound Algorithm V2.3. Initial focus areas are anticipated to include mitigating subtle pre-echo artifacts identified in preliminary internal listening sessions and enhancing the perceived vertical soundstage resolution. The refinement process will leverage advanced adaptive HRTF interpolation techniques and dynamic room modeling parameters to achieve these objectives. Furthermore, to ensure long-term scalability, optimize power efficiency, and potentially reduce the Bill of Materials (BOM) cost, the Hardware Integration Team will initiate a detailed exploration of advanced DSP hardware platforms. This will include a comparative analysis of the Cadence Tensilica HiFi 5 and CEVA-XM6 architectures, alongside a preliminary feasibility study for a custom GeniCo Application-Specific Integrated Circuit (ASIC) solution for future product generations. Evaluation criteria will encompass computational throughput (GFLOPS/W), memory bandwidth, I/O flexibility, and the maturity of associated development toolchains.

The upcoming Sprint 7, scheduled from August 1 to August 14, 2026, will strategically allocate engineering and research resources to execute these immediate next steps. The following table delineates specific team assignments, primary responsibilities, and expected deliverables for this critical two-week cycle, ensuring clear accountability and

focused effort across all project facets.

Team / Lead	Primary Assignments	Key Deliverables (Sprint 7: Aug 1 - Aug 14, 2026)	
Category	Specific Action / Task	Lead / Team	Completion
A. Algorithm Testing & Refinement			
A.1. Human Subject A/B Testing	Finalize A/B test protocol and participant recruitment.	Dr. E. Reed (Research)	2026-07-25
	Conduct initial A/B test sessions (Phase 1: Internal GeniCo users).	Dr. A. Sharma (Algorithm)	2026-08-05
	Conduct A/B test sessions (Phase 2: External audiophile volunteers).	Dr. E. Reed (Research)	2026-08-12
	Collect and preprocess subjective preference scores and qualitative feedback.	Dr. A. Sharma (Algorithm)	2026-08-14
A.2. Algorithm Refinement (V2.3)	Analyze initial A/B test data to identify primary areas for algorithmic improvement.	Dr. A. Sharma (Algorithm)	2026-08-16
	Develop and implement initial code iterations for AuraSound Algorithm V2.3.	Algorithm Dev Team	2026-08-23
	Conduct internal listening tests of Algorithm V2.3 prototypes.	Dr. E. Reed (Research)	2026-08-28
B. DSP Hardware Exploration			
B.1. Platform Evaluation	Complete detailed technical specifications review for Cadence Tensilica HiFi 5 and CEVA-XM6.	Eng. M. Jensen (Hardware)	2026-08-07
	Initiate preliminary discussions with vendor representatives for evaluation kits and support.	Eng. M. Jensen (Hardware)	2026-08-09
	Conduct a preliminary feasibility study for a custom GeniCo ASIC solution.	Hardware Integration Team	2026-08-18
B.2. Integration & Benchmarking	Prepare test benches and development environments for selected DSP evaluation boards.	Hardware Integration Team	2026-08-21
	Conduct initial benchmarking of AuraSound V2.2 on prospective DSP platforms.	Eng. M. Jensen (Hardware)	2026-08-29
C. Resource Allocation & Planning			
C.1. Sprint 7 Review	Conduct end-of-sprint review meeting to assess progress and deliverables.	Dr. E. Reed (Research)	2026-08-14

	Plan Sprint 8 objectives based on A/B test insights and DSP exploration.	Project Core Team	2026-08-20
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